

Part II

Understanding People

Introduction to understanding people

Can we design for people if we do not understand them? History has shown that we cannot—user interfaces that have failed often lacked awareness of users’ behaviors, feelings, needs, and wants. For example, some mobile text entry methods have failed because they did not support users in becoming better at using them [436], and some collaborative systems have failed because they did not incorporate an understanding of people’s needs, motivations, and work practices [302].

In the introductory part of this book (Part I), it was noted that human-centeredness is one of the pillars of human–computer interaction (HCI) as a discipline. It requires us to take a real interest in people. It requires us to base our decisions in research and design on our understanding of people. But *how* is one human-centered—what does it mean to work in a human-centered way?

To claim that a choice is made with people in mind—that it is *human-centered*—means that the choice is justified by reference to knowledge of how people feel, think, and behave. For example, we may choose colors for a user interface based on how the human visual system works, design mechanisms in an application to motivate changes in unwanted behaviors such as smoking, or choose to develop a product concept based on ideas about what people desire. Furthermore, decisions about how to evaluate a collaborative system can be based on knowledge about how people coordinate work when aiming to achieve a shared goal.

In these and many other instances, we know things about people—from research in HCI or through other sciences—and put that knowledge to use in HCI activities. We call this knowledge and its use to improve interactive systems *understanding people*.

Understanding people is hard. Most of the factors influencing a person’s observable behavior are hidden. We cannot precisely know what a person thinks, what they feel, or what drives a person’s behavior. However, using scientific methods and theories, it is nonetheless possible to obtain some knowledge of these factors. Such knowledge gathering is valuable in HCI because it allows us to reason about user behavior.

To illustrate such reasoning, compare the two photos in Figure 2.1.

Perception: The two situations differ in terms of what the users can perceive. During driving, the user *shares* their visual attention between the computer and the road, where the latter is continuously changing. In many countries, using a mobile device while driving is illegal because of the effect of the device on the user’s ability to perceive events on the road.

Motor control: Many games require not only fast reflexes but also the ability to intercept a fast-moving target that can be small and move erratically. Using a mobile device for non-gaming purposes mostly involves static targets selected with a very different input device, such as a touchscreen.

Thinking: The gamer must keep track of several events, such as the status of enemy players, to choose their next action effectively. By contrast, the selection of options from a mobile display is based on the recognition of icons and labels that are most likely to lead to the target state.

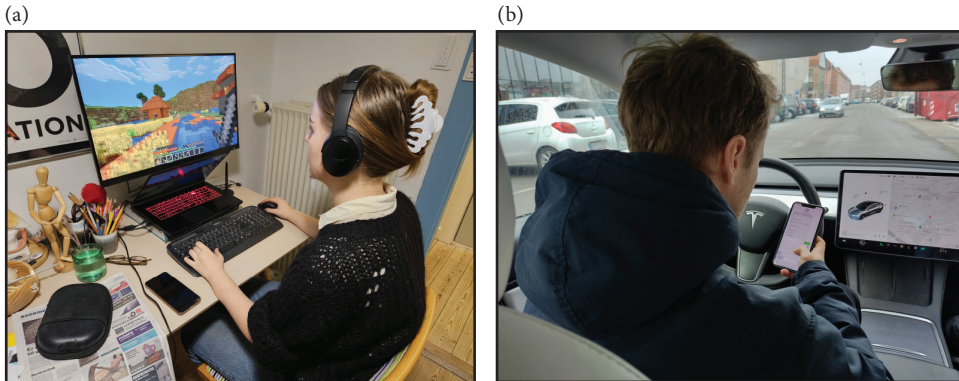


Figure 2.1 Chapter 2 introduces methods and theories that help understand people as users of computers. Compare the two situations in the photos: desktop gaming and driving. How do you believe they differ in terms of the users' goals, experience, thinking, and cognitive demands?

Needs: What kinds of desires and risks are associated with the two situations? Do people play games for different reasons than those that drive the use of social media? Or are there a few basic psychological needs that can be met equally well through either activity?

Experience: Immersion in a virtual world is a desired quality in gaming but not in mobile interaction, which places greater emphasis on instrumental experiences such as being able to complete tasks.

Collaboration: In the game example, the ad hoc formation of a team of players and their mutual awareness of what is going on are essential to the team's success. Understanding how to achieve this collaborative state is an important part of developing interactive systems. Driving a car, by contrast, requires a different type of collaboration, for instance, with any other people in the car and other road users.

Communication: Competitive gaming is an extreme situation in which two parties have conflicting goals. Both users must infer what the other party means or desires via the limited cues on the interface. Every so often, this requires intense communication and impromptu collaboration. Similarly, driving, in this case, co-occurs with the use of a mobile phone, which shapes what is being communicated by the multitasking user.

We write about people in other parts of the book. What is the difference between this part and the others? The defining focus of this part is the pursuit of *general* knowledge about people, for instance, as theories, models, concepts, and taxonomies. We seek to understand how people feel, think, and behave in ways that hold across different times and settings in computer use. This is a very important type of understanding because it has wide applicability. For example, the mechanisms involved in how the human visual system works are universally relevant for graphical user interfaces.

At the same time, such knowledge about people is *abstract*: It does not prescribe a *specific* user, setting, or configuration. Therefore, to be valuable for design and engineering, the understanding described in this part of the book needs to be complemented by other forms of insights about users. Table 2.1 shows an overview of the types of knowledge covered in this book. The part on user research (Part III) covers methods for gathering information on specific users, activities,

Table 2.1 Different types of knowledge about people in interaction with computing systems.

Topic	Part	Focus
Understanding people	This part	Generalizable understanding of how people think, feel, and behave
User research	Part III	Methods for obtaining insights on specific users, activities, and use contexts
Interaction	Part V	Generalizable understanding of how users interact with computer systems
Evaluation	Part VIII	Methods for assessing interaction with specific interactive systems

and contexts of use. The part on interaction (Part IV) provides HCI theories and models that explain how users interact with a variety of computer systems. Finally, the part on evaluation (Part VIII) explains how to assess HCI systems using a variety of different approaches, such as experiments and deployment studies.

Understanding people is central to HCI, and many HCI studies contribute new insights that can inform design. Next, we provide some examples.

- Borst et al. [88] proposed and evaluated a process model to predict when interruptions are disruptive to users. Their experiments support two design guidelines to minimize disruptions resulting from interruptions.
- Bachynskyi et al. [34] studied user performance and ergonomics on different touch surfaces, such as public displays, tabletops, tablets, and smartphones, thereby informing the design of user interfaces that use these surfaces for interaction.
- Lottridge et al. [495] investigated how chronic multitasking with relevant and irrelevant distractors affects the quality of writing. They found that multitaskers write better essays when provided with relevant distractors and worse essays when provided with irrelevant distractors. This tells us that while multitaskers can be negatively affected by irrelevant distractors, they are also able to integrate different sources of information when writing essays.

Next, we review the different forms of understanding we employ in HCI, the different areas of understanding, and how we use them in evaluation and design.

2.1 Types of understanding

Several types of understanding are relevant in HCI. A concept may help us to see an underlying driving factor, such as a need or motivation, behind users' behavior. A computational model enables us to predict what a user can recall. A model may help us separate the actions that a user performs when searching for information. What is common to the types of understanding covered in this part, and to those just mentioned, is that they are general, robust across time and place, and linkable to empirical phenomena. In this part, we will discuss five main types of understanding.

Theories: They consist of constructs and relations among those constructs. Theories help us understand, explain, or predict phenomena related to interactive systems. They are more

general and encompassing than models. For example, in Chapter 6, we describe self-determination theory. This theory outlines the motivations of people. It describes the factors involved in intrinsic motivation and the general human tendency to engage in activities that are seen as enjoyable and interesting.

Concepts: They name particular phenomena, often with additional characteristics, such as how to identify the phenomena, information on when they usually occur, or knowledge about their underpinning mechanisms. For example, in Chapter 9, we learn about turn-taking as a concept for understanding human conversations offline and online. This concept has implications for how we support communication in interactive systems.

Taxonomies: They propose a system of elements or mechanisms of how people think, feel, or act. For example, we describe how human memory consists of several systems in Chapter 5. A simple taxonomy separates declarative and procedural memory, each broken down into further, distinct types of knowledge. This taxonomy may be used to analyze the types of knowledge that a particular interface requires.

Models: Models are formally expressed simplifications of reality. They link concepts, often in a visual form, but also mathematically or through computer code. They may be verbal or quantitative, provide numerical estimates, or allow the model to be simulated computationally. For example, Chapter 4 discusses models that predict how design affects the time it takes people to select a target, such as an icon, on a display.

Guidelines: Guidelines are practical rules of thumb or heuristics that summarize theoretical knowledge. For example, the chapter on cognition (Chapter 5) provides knowledge from which one can derive guidelines for evaluating interactive systems (Chapter 41).

These five types of understanding represent different forms of understanding people. However, such understanding cannot be static. In HCI, effective understanding has four qualities. First, researchers can and should update their understanding in light of new observations. An understanding of people that is immutable is not understanding—it is a dogma. Research progresses by advancing or rejecting previous knowledge based on new evidence.

Second, simplification is often the crux of understanding—not just in HCI but also in the social and behavioral sciences in general. Something complex, such as someone's behavior, is explained by reference to something more straightforward, such as a construct like personality, or a concept about collaboration.

Third, understanding people should be actionable to be useful in HCI. Knowledge should help practitioners make better choices. An understanding of people that only calls for more empirical research may be interesting to academic researchers looking for a new line of inquiry but ultimately has little practical value in itself. Thus, our discussions in this part of the book balance an extensive theoretical understanding of some aspects of people with simple—actionable—approximations.

Fourth, claims about people should be logically sound and empirically justifiable. Theories are subject to constant scrutiny and public criticism. A theory based on someone's opinion is not really a theory. Scrutiny can take place both in a related discipline, such as psychology, and in the field of HCI. In the construction of theories to understand users, HCI has drawn from several areas of psychology, most notably cognitive psychology, social psychology, and the psychology of motivation and needs. The application of such theories in HCI has frequently resulted in HCI research feeding information back to the original disciplines that provided the theories.

2.2 Areas of understanding

HCI is fascinating because it encompasses a wide range of human activities, from delicate adaptations of movement in virtual reality to changes in adolescents' well-being due to digital communication technologies. It includes people browsing a web page and local communities using social media to organize activities. The areas of understanding relevant to HCI therefore include *all* areas of understanding people and encompass many areas of scholarship. However, which of these areas are the most beneficial to learn about?

2.2.1 Seven areas of understanding people

In this book, we focus on seven areas that have historically been prominent in HCI research and that cover many types of understanding needed in practice. This part of the book, Part II, summarizes what each area teaches us about the individual and social factors that affect the use of computers. Table 2.2 shows an overview of these areas.

These areas allow us to understand the basic perception and motor aspects of interaction, making us aware of the limits of human performance and how design can shape such performance. The areas help us discuss people's cognition and how their needs shape people's behavior and their experiences with interactive systems. They also help us understand communication and collaboration. The areas mentioned in Table 2.2 are mainly rooted in sociology and social psychology. Knowing these areas is essential for the design of applications and services that are used by groups and organizations.

All of these areas are important and interdependent. Independently of the interactive system and the people involved, any task will involve motor control. Furthermore, all people form experiences

Table 2.2 Seven areas of understanding people that provide useful knowledge on the human use of computing.

Area	Focus	Relevance for design
Perception	How people see, feel, hear, taste, and smell	Informs the design of user interfaces
Motor control	How people plan and execute movements	Informs the design of input devices and interaction techniques
Cognition	How people remember, pay attention, and think	Informs the design of complex interactive tasks that require memory and reasoning
Needs	How needs motivate people	Tells us what is important for users in computer use and beyond
Experience	How people experience and form experiences	Tells us how users experience events involving computers
Communication	How people communicate with each other	Informs the design of services and applications for human–human communication
Collaboration	How people achieve joint goals	Informs the design of collaborative software

along the lines outlined later when they interact with systems. We, therefore, reject claims that some of these areas are of low priority for understanding people and that some of these areas have faded in importance for HCI. Nevertheless, specific projects or interactive systems may require particular attention to certain areas. For example, a study of sharing bereavement on social media may require particular attention to the areas of communication (Chapter 9) and experience (Chapter 7). If we are interested in creating a new way of working together in virtual reality, we need to know about both perception (Chapter 3) and collaboration (Chapter 8). However, focusing on specific areas does not mean the other areas are irrelevant.

2.2.2 Special application areas

Special application areas are specific areas of activity, such as work or games, whose characteristics make them unique. These areas are unique in terms of what people do, how they feel, or what they think to such a large extent that it is invaluable to study them. Research in such areas often involves both general theories and principles and specific ones that are developed especially for each area.

The special areas that HCI engages with are continuously changing. Recent examples include personal health, leisure, sustainability, technology in the developing world, creativity, games, and learning. Much can be learned about such areas through user research, following the principles outlined in Part III. The scientific literature outside HCI is likely to contain a lot of general information as well. For example, there is a lot of knowledge in the literature about why people play games. This includes theories of motivation in games, particular behaviors in games, and measures of states in games, such as immersion or flow (Chapter 7). In studies of health and well-being, we employ theories of behavior change and motivation (Chapter 6). In studies of security and privacy, we consider users' basic needs for personal safety (Chapter 6). In studies of computer-supported cooperative work, we employ an understanding of how people collaborate (Chapter 8) and communicate (Chapter 9). A proper understanding of such special application areas, regardless of how particular they are, should always include general knowledge of HCI.

2.2.3 User groups

Another area of general understanding concerns *user groups*. User groups are differentiated by their interests, capabilities, and the systems they use. They may differ in important ways across the seven areas of understanding people. Therefore, HCI studies, as well as other fields, often take into account the characteristics of user groups. One example of a user group is older adults. Much is understood about this user group in terms of sensory, motor, and cognitive capabilities, as well as disabilities. Such understanding can be critical for projects concerning older adults. Numerous other user groups have also attracted a substantial amount of research, such as design and evaluation activities involving children, users with disabilities, people in low-income countries, and families.

2.2.4 Individual differences

It is almost a cliché that everyone is different; in addition to looking at how groups of users differ, we also need to understand how users' characteristics vary. No matter how homogeneous a user sample is, significant differences between individuals are the norm, not an exception [241].

Individual differences can also be pronounced in HCI tasks. In tasks such as text editing and programming, differences between individuals can be in the order of 20:1 [195]. The scale of such differences can have great practical significance.

To go beyond merely stating that differences exist, we need to understand the mechanisms that produce them. Why is age, gender, or education an important factor? In this part of the book, we will learn about the mechanisms that underpin differences among users.

2.3 Applying our understanding of people

The HCI field values *actionable* knowledge. However, how can we apply our knowledge about people? The general answer to this question is that understanding people *helps solve problems in HCI*. Many research problems in HCI involve explaining and predicting how people use computers, considering what systems to construct, or investigating how an interactive system affects a community or organization. The theories presented in this part of the book help tackle such problems.

Work in HCI has explained how we might apply theory in this field. For instance, Bederson and Shneiderman [56] and Rogers [706] separated descriptive, explanatory, predictive, prescriptive, and generative theories in HCI. We use this categorization to identify some ways of using different understandings of people and present some examples of such uses.

First, we may use an understanding of people *to direct what to pay attention to* in a project. Understanding people in general might help direct attention in encounters with particular users. For example, research shows that motivation plays a key role in what people do and what value they derive from their actions (Chapter 6). Thus, in encounters with users in a particular project, we may want to pay attention to their motivations. We may also use such knowledge about people when we analyze and reflect on such encounters.

Second, we may use the understanding of people *to explain empirical findings*. For example, users may mention the need to stay in touch during a particular activity. This may be explained by the concept of coordination work (Chapter 8). As another example, Steinberger et al. [793] explored how to increase engagement in driving via gamification. They constructed prototypes and explained their empirical results on driving with a general model of opportunities and costs.

Third, we may use the understanding of people *to make design decisions*. These may be high-level decisions concerning the overall concept of an interactive system. For instance, we may depart from the typical biases of memory (Chapter 5) to design a photo-based memory tool [345]. As another example, Consolvo et al. [167] used theories of behavior change to build a system that encourages people to be more physically active. Theories may also inform particular design decisions. Oulasvirta et al. [629] used models of how perception is integrated across different channels (Chapter 3) to discuss the design of buttons.

Fourth, we may use an understanding of people to help *explore a design space*. That way, general understanding, in addition to direct contact with users (see the discussion on user research in Part III), serves to generate design ideas. For example, Ballendat et al. [37] used proxemics—a theory of the distances people perceive and how people use physical distance in encounters with other people—to generate design ideas. They created a suite of devices and interaction techniques based on an awareness of nearby people and other devices. In short, the theory of proxemics served an important generative role.

Fifth, we may use an understanding of people to *predict people's behavior*. Predictions may refer to expected events or processes, and they may involve numerical estimates. For example, researchers have modeled visual search patterns on web pages. Such models allow us to predict the most salient part of a web site with some probability, or predict the average time needed to find a particular link on a web page.

2.4 Is a general understanding of people possible?

We have argued that there are general understandings of people that are useful across topics in HCI. We have argued that such understandings are useful for solving practical problems in HCI, although a consequence of their generality is that they must be augmented with user research.

However, this view is not accepted by all. Some researchers argue that a general understanding of people is impossible. The argument is that general understanding is not typical of any particular situation; as such, it does not provide much, or any, benefit. Furthermore, Lincoln and Guba [483] argued that generalizations imply a belief in determinism—if the antecedent of a generalization is present, the consequent must follow—and some form of reductionism—phenomena relate only to one or a few generalizations. While we agree that it is difficult to arrive at generalized knowledge, we argue that such understandings exist in many areas and show examples of how they may be used. For example, we know things about the human visual system and collaboration that can—and indeed must—play a role in designing and evaluating interactive systems.

Another related belief is that we are all equipped with empathy—the ability to intuitively understand other people and their experiences. Why do we need theories if we have such an ability? Unlike empathy, theories can be communicated to others and subjected to scrutiny. Moreover, our empathy and intuition often fail us. The inferences we make about other people are often incorrect, especially when they involve people from different backgrounds (Chapter 5). Scientific concepts are more comprehensive, precise, and—if properly applied—appropriate for describing what happens when a user interacts with a system.

Finally, why bother to understand people in the first place? Why not just create a quick prototype of an interactive system and improve the design later on? Indeed, Petroski [652] described the evolution of everyday artifacts as occurring through trial and error. The pencil, for example, has gone through hundreds of development cycles throughout its history, some related to the manufacturing of pencils, and some to its use. In a discussion about the role of theory in HCI, Landauer [453] argued that we need to “get real” about the possible impact of cognitive psychology—a type of understanding of people—in HCI. The argument is that the real world of users is not captured well or in its entirety by cognitive psychology. Instead, empirical work and formative evaluation are suggested to be sufficient to drive the development of interactive computing systems. That is, rather than using an understanding of people, we rely on empirically testing user interfaces with people. We share this empirical orientation. However, relying exclusively on empirical data is costly and prone to failure.

In summary, we find theories about people essential. Although such theories need to be complemented by empirical research, we think that the HCI field can, and should, draw on general understandings of people. The rest of this part presents such understandings.

Summary

- A deep and scientific engagement with interactive systems and the phenomena that surround them begins with people. We call this approach human-centered.
- Being human-centered requires us to draw on theories and models of how people feel, think, and behave.
- There are seven key areas of understanding people: (1) perception, (2) motor control, (3) cognition, (4) needs, (5) experience, (6) communication, and (7) collaboration.
- These understandings of people are general, holding across many individuals, many types of user interfaces, and many use contexts.